HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2013)

Student No.: 13 Name: _____ Marks: / 38

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. If $\frac{y-1}{c} = \frac{y+1}{d}$, then $y =$

   A. $\frac{c-d}{c+d}$

   B. $\frac{d-c}{c+d}$

   C. $\frac{c+d}{c-d}$

   D. $\frac{c+d}{d-c}$

(DSE-2013 PII Q02)

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2. $0.0504545 =$

A. 0.051 (correct to 2 significant figures)

B. 0.0505 (correct to 3 decimal places)

C. 0.05045 (correct to 4 significant figures)

D. 0.05046 (correct to 5 decimal places)

(DSE-2013 PII Q04)

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3. The solution of $x - \frac{x-1}{2} > 5$ or $1 < x - 11$ is

A. $x > 9$

B. $x > 10$

C. $x > 11$

D. $x > 12$

(DSE-2013 PII Q05)

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4. Susan sells two cars for $80 080 each. She gains 30% on one and loses 30% on the other. After the transactions, Susan

   A. loses $15 840.

   B. gains $5 544.

   C. gains $10 296.

   D. has no gain and no loss.

(DSE-2013 PII Q10)

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5. The actual area of a playground is $900\,m^2$. Its area on a map is $36\,cm^2$. Then the scale of the map is

A. 1 : 25

B. 1 : 50

C. 1 : 500

D. 1 : 250 000

(DSE-2013 PII Q12)

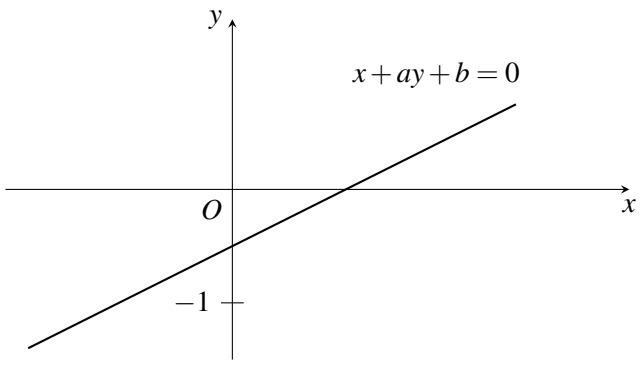
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6. The figure shows $x+ay+b=0$. Which statements are true?

I. $a<0$ II. $b<0$ III. $a<b$



A. I and II only

B. I and III only

C. II and III only

D. I, II and III

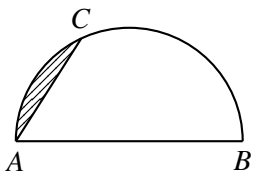
(DSE-2013 PII Q14)

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7. In the figure, the diameter of the semicircle ABC is 3 cm. If chord $AC = 2$ cm, find the area of the shaded region corrected to the nearest $0.01\,\text{cm}^2$.



A. $0.23\,\text{cm}^2$

B. $0.52\,\text{cm}^2$

C. $0.64\,\text{cm}^2$

D. $1.07\,\text{cm}^2$

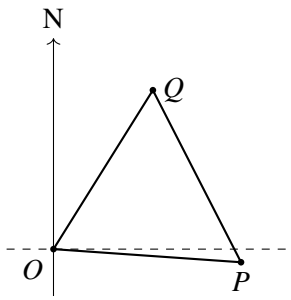
(DSE-2013 PII Q16)

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8. In the figure, the bearing of P from O is $S86^\circ E$ and the bearing of from O is $N32^\circ E$. If P and Q are equidistant from O, then the bearing of P from Q is



A. N24°W

B. N27°W

C. S24°E

D. S27°E

(DSE-2013 PII Q20)

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9. The equation of the circle C is $2x^2+2y^2-4x+8y-5=0$. The coordinates of the points P and Q are $(-1,2)$ and $(4,0)$ respectively. Which of the following is/are true?

I. The radius of C is 5.

II. The midpoint of PQ lies outside C.

III. If G is the centre of C, then $\angle PGQ$ is acute angle.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2013 PII Q25)

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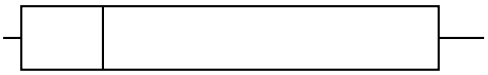
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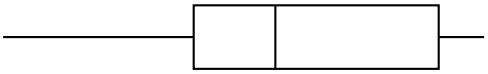
10. The stem-and-leaf diagram shows hourly wages. Which box-and-whisker diagram may represent them?

4	0 2 2 2 4 4 4 7
5	0 0 1 2 2 6 8 9
6	3 5 5 7
7	0
8	2 6
9	5

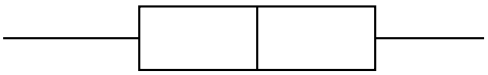
A.



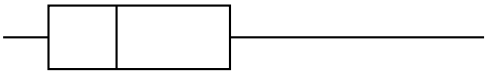
B.



C.



D.



(DSE-2013 PII Q29)

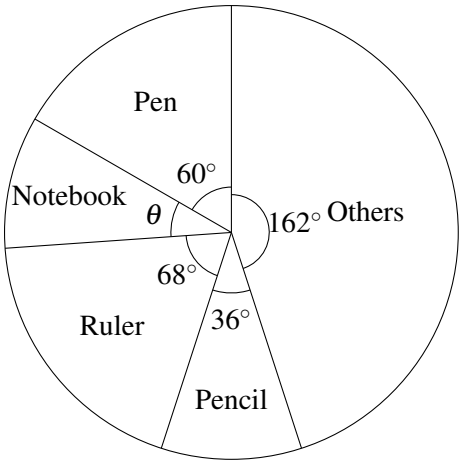
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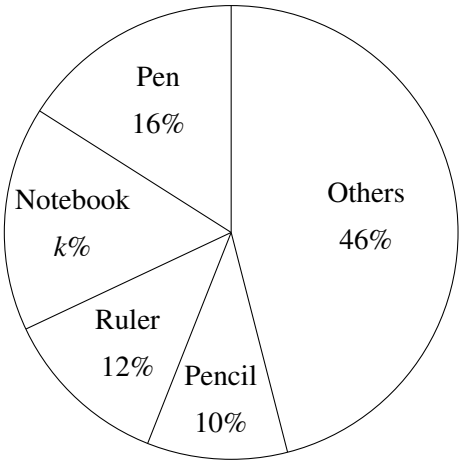
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11. The pie charts below show the distributions of the profits of stationery shop$X$ and stationery shop Y from the sales of stationery in a certain month. Which of the following must be true?

Distribution of the profits of stationery shop X



Distribution of the profits of stationery shop Y



A. In that month, the profit from the sales of pencils of stationery shop X is the same as that of stationery shop Y.

B. In that month, the total profit from the sales of pens and notebooks of stationery shop X is less than the total profit from the sales of rulers and pencils of the shop.

C. $k = 14$

D. $\theta = 36^\circ$

(DSE-2013 PII Q30)

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12. The L.C.M. of a^2+4a+4, a^2-4 and a^3+8 is

A. $a+2$

B. $(a-2)(a+2)^2(a^2-2a+4)$

C. $(a-2)(a+2)^2(a^2+2a+4)$

D. $(a-2)(a+2)^4(a^2-2a+4)$

(DSE-2013 PII Q31)

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13. $\text{A00000E00001}_{16} =$

A. $10 \times 16^{11} + 14 \times 16^5 + 17$

B. $11 \times 16^{11} + 15 \times 16^5 + 17$

C. $10 \times 16^{12} + 14 \times 16^6 + 272$

D. $11 \times 16^{12} + 15 \times 16^6 + 272$

(DSE-2013 PII Q33)

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14. If $x - \log y = x^2 - \log y^2 - 10 = 2$, then $y =$

A. 100

B. 2 or -4

C. $\frac{1}{100}$ or 10000

D. $\frac{1}{10000}$ or 100

(DSE-2013 PII Q34)

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15. The nth term of a sequence is $2n - 19$. Which are true?

I. 25 is a term of the sequence..

II. The sequence has 10 negative terms.

III. The sum of the first n terms of the sequence is $n^2 - 18n$.

A. I only

B. II only

C. I and III only

D. II and III only

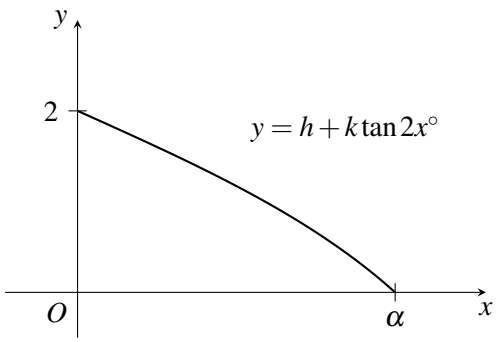
(DSE-2013 PII Q38)

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16. Let h and k be constants. The figure shows the graph of $y = h + k\tan 2x^\circ$, where $0 \le x \le \alpha$. Which of the following are true?



I. $h > 0$

II. $k < 0$

III. $\tan\alpha = \frac{1}{k}$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2013 PII Q39)

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17. If the height of a regular tetrahedron has height 2 cm. then the volume of the tetrahedron is

A. $2\,\text{cm}^3$

B. $\sqrt{3}\,\text{cm}^3$

C. $\sqrt{6}\,\text{cm}^3$

D. $3\sqrt{3}\,\text{cm}^3$

(DSE-2013 PII Q40)

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18. Find the range of k such that the circle $x^2 + y^2 + 2x - 2y - 7 = 0$ and the straight line $3x - 4y + k = 0$ intersect.

A. $-8 < k < 22$

B. $-8 \leq k \leq 22$

C. $k < -22$ or $k > 8$

D. $k \leq -22$ or $k \geq 8$

(DSE-2013 PII Q42)

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19. If the variance of the five numbers x_1, x_2, x_3, x_4 and x_5 is 13, then the variance of the five numbers$3x_1 + 4, 3x_2 + 4, 3x_3 + 4, 3x_4 + 4$ and $3x_5 + 4$ is

A. 39

B. 43

C. 117

D. 121

(DSE-2013 PII Q45)

I can finish _____ questions

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